

Central limit theorem for multi-type general branching processes

Alicja Kołodziejska

JLU Gießen / University of Wrocław

Będlewo, May 2026

Notation and Assumptions

Recall:

- $p \in \mathbb{N}$ – number of types; $[p] = \{1, \dots, p\}$;
- ξ – the reproduction process with intensity measure μ ;
- $\mathcal{L}\mu$ – Laplace transform of μ .

Notation and Assumptions

Recall:

- $p \in \mathbb{N}$ – number of types; $[p] = \{1, \dots, p\}$;
- ξ – the reproduction process with intensity measure μ ;
- $\mathcal{L}\mu$ – Laplace transform of μ .

We assume that

- $\mathcal{L}\mu(0)$ is irreducible
 $\rightsquigarrow$ we see all the types
- there exists the Malthusian parameter $\alpha > 0$:
 $\mathcal{L}\mu(\alpha)$ has Perron-Frobenius eigenvalue 1
 $\rightsquigarrow$ growth of rate $e^{\alpha t}$

The CMJ process

For a (random) characteristic $\varphi : \mathbb{R} \rightarrow \mathbb{R}^p$,

$$Z_t^\varphi = \sum_u \mathbf{e}_{\tau(u)}^\top \varphi_u(t - S(u)).$$

τ — type
 S — time of birth

The CMJ process

For a (random) characteristic $\varphi : \mathbb{R} \rightarrow \mathbb{R}^p$,

$$\mathcal{Z}_t^\varphi = \sum_u \mathbf{e}_{\tau(u)}^\top \varphi_u(t - S(u)).$$

τ — type
 S — time of birth

Example: If $\varphi(t) = \mathbf{a} \mathbb{1}_{[0,\infty)}(t)$ for some $\mathbf{a} \in \mathbb{R}^p$, then

$$\mathcal{Z}_t^\varphi = \sum_u \mathbf{e}_{\tau(u)}^\top \mathbf{a} \mathbb{1}_{[0,\infty)}(t - S(u))$$

The CMJ process

For a (random) characteristic $\varphi : \mathbb{R} \rightarrow \mathbb{R}^p$,

$$\mathcal{Z}_t^\varphi = \sum_u \mathbf{e}_{\tau(u)}^\top \varphi_u(t - S(u)).$$

τ — type
 S — time of birth

Example: If $\varphi(t) = \mathbf{a} \mathbb{1}_{[0,\infty)}(t)$ for some $\mathbf{a} \in \mathbb{R}^p$, then

$$\mathcal{Z}_t^\varphi = \sum_u \mathbf{e}_{\tau(u)}^\top \mathbf{a} \mathbb{1}_{[0,\infty)}(t - S(u)) = \sum_u \mathbf{a}_{\tau(u)} \mathbb{1}_{S(u) \leq t}$$

counts the number of particles born up to time t , weighted by type according to $\mathbf{a}$.

Law of large numbers

In our setting [Nerman's LLN],

$$e^{-\alpha t} \mathcal{Z}_t^\varphi \xrightarrow{t \rightarrow \infty} \frac{W}{\beta} \int_{\mathbb{R}} \mathbf{u}^\top \mathbb{E}[\varphi](s) e^{-\alpha s} \mathrm{d}s \quad \mathbb{P}\text{-a.s.},$$

Law of large numbers

In our setting [Nerman's LLN],

$$e^{-\alpha t} \mathcal{Z}_t^\varphi \xrightarrow{t \rightarrow \infty} \frac{W}{\beta} \int_{\mathbb{R}} \mathbf{u}^\top \mathbb{E}[\varphi](s) e^{-\alpha s} ds \quad \mathbb{P}\text{-a.s.},$$

where

- $\mathbf{v}, \mathbf{u}$ – right and left eigenvectors of $\mathcal{L}\boldsymbol{\mu}(\alpha)$ such that $\|\mathbf{v}\|_1 = 1, \mathbf{u}^\top \mathbf{v} = 1$;
- $\beta = \mathbf{u}^\top (-(\mathcal{L}\boldsymbol{\mu})'(\alpha)) \mathbf{v}$;
- W is the limit of [Nerman's martingale](#)

$$W_t = \sum_{u \in \mathcal{C}_t} \mathbf{v}_{\tau(u)} e^{-\alpha S(u)}.$$

The mean and the martingales

$$\mathbb{E}[\mathcal{Z}_t^\varphi] = \sum_{\lambda \in \Lambda} e^{\lambda t} \sum_{k=0}^{k_\lambda-1} A_{\lambda,k+1} \int_{\mathbb{R}} \mathbb{E}[\varphi](s) \frac{(t-s)^k}{k!} e^{-\lambda s} \mathrm{d}s + r(t),$$

The mean and the martingales

$$\mathbb{E}[\mathcal{Z}_t^\varphi] = \sum_{\lambda \in \Lambda} e^{\lambda t} \sum_{k=0}^{k_\lambda-1} A_{\lambda, k+1} \int_{\mathbb{R}} \mathbb{E}[\varphi](s) \frac{(t-s)^k}{k!} e^{-\lambda s} ds + r(t),$$

where

$$\Lambda = \{\lambda \in \mathbb{C} : \det(I - \mathcal{L}\boldsymbol{\mu}(\lambda)) = 0, \operatorname{Re}\lambda \in [\alpha/2, \alpha]\},$$
$$(I - \mathcal{L}\boldsymbol{\mu}(z))^{-1} = (z - \lambda)^{-k_\lambda} A_{\lambda, k_\lambda} + \cdots + (z - \lambda)^{-1} A_{\lambda, 1} + R_\lambda(z).$$

The mean and the martingales

$$\mathbb{E}[\mathcal{Z}_t^\varphi] = \sum_{\lambda \in \Lambda} e^{\lambda t} \sum_{k=0}^{k_\lambda-1} A_{\lambda,k+1} \int_{\mathbb{R}} \mathbb{E}[\varphi](s) \frac{(t-s)^k}{k!} e^{-\lambda s} \mathrm{d}s + r(t),$$

where

$$A_{\lambda,k} = \mathbb{E}[W_t(\lambda, k)]$$

for the martingale

$$W_t(\lambda, k) = \sum_{u \in \mathcal{C}_t} e^{-\lambda S(u)} \sum_{j=k}^{k_\lambda} \frac{(-S(u))^{j-k}}{(j-k)!} \mathbf{e}_{\tau(u)}^\top A_{\lambda,j}.$$

The mean and the martingales

$$\mathbb{E}[\mathcal{Z}_t^\varphi] = \sum_{\lambda \in \Lambda} e^{\lambda t} \sum_{k=0}^{k_\lambda-1} A_{\lambda,k+1} \int_{\mathbb{R}} \mathbb{E}[\varphi](s) \frac{(t-s)^k}{k!} e^{-\lambda s} ds + r(t),$$

where

$$A_{\lambda,k} = \mathbb{E}[W_t(\lambda, k)]$$

for the martingale

$$W_t(\lambda, k) = \sum_{u \in \mathcal{C}_t} e^{-\lambda S(u)} \sum_{j=k}^{k_\lambda} \frac{(-S(u))^{j-k}}{(j-k)!} \mathbf{e}_{\tau(u)}^\top A_{\lambda,j}.$$

If $\operatorname{Re} \lambda > \frac{\alpha}{2}$, under second moment assumptions on $\boldsymbol{\xi}$,

$$W_t(\lambda, k) \xrightarrow{L^2} W(\lambda, k), \quad A_{\lambda,k} = \mathbb{E}[W(\lambda, k)].$$

Asymptotic behaviour when $\partial\Lambda = \emptyset$

We expect, if $\operatorname{Re}\lambda > \frac{\alpha}{2}$ for all $\lambda \in \Lambda$,

$$\mathcal{Z}_t^\varphi \approx \sum_{\lambda \in \Lambda} e^{\lambda t} \sum_{k=0}^{k_\lambda-1} W(\lambda, k+1) \int_{\mathbb{R}} \mathbb{E}[\varphi](s) \frac{(t-s)^k}{k!} e^{-\lambda s} \mathrm{d}s + o(e^{\frac{\alpha}{2}t}).$$

Asymptotic behaviour when $\partial\Lambda = \emptyset$

We expect, if $\operatorname{Re}\lambda > \frac{\alpha}{2}$ for all $\lambda \in \Lambda$,

$$\mathcal{Z}_t^\varphi \approx \sum_{\lambda \in \Lambda} e^{\lambda t} \sum_{k=0}^{k_\lambda-1} W(\lambda, k+1) \int_{\mathbb{R}} \mathbb{E}[\varphi](s) \frac{(t-s)^k}{k!} e^{-\lambda s} \mathrm{d}s + o(e^{\frac{\alpha}{2}t}).$$

One may check that $k_\alpha = 1$ and

$$A_{\alpha,1} = \frac{1}{\beta} \mathbf{v} \mathbf{u}^\top$$

Asymptotic behaviour when $\partial\Lambda = \emptyset$

We expect, if $\operatorname{Re}\lambda > \frac{\alpha}{2}$ for all $\lambda \in \Lambda$,

$$\mathcal{Z}_t^\varphi \approx \sum_{\lambda \in \Lambda} e^{\lambda t} \sum_{k=0}^{k_\lambda-1} W(\lambda, k+1) \int_{\mathbb{R}} \mathbb{E}[\varphi](s) \frac{(t-s)^k}{k!} e^{-\lambda s} \mathrm{d}s + o(e^{\frac{\alpha}{2}t}).$$

One may check that $k_\alpha = 1$ and

$$A_{\alpha,1} = \frac{1}{\beta} \mathbf{v} \mathbf{u}^\top, \quad \text{so} \quad W(\alpha, 1) = \frac{W}{\beta} \mathbf{u}^\top.$$

Asymptotic behaviour when $\partial\Lambda = \emptyset$

We expect, if $\operatorname{Re}\lambda > \frac{\alpha}{2}$ for all $\lambda \in \Lambda$,

$$\mathcal{Z}_t^\varphi \approx \sum_{\lambda \in \Lambda} e^{\lambda t} \sum_{k=0}^{k_\lambda-1} W(\lambda, k+1) \int_{\mathbb{R}} \mathbb{E}[\varphi](s) \frac{(t-s)^k}{k!} e^{-\lambda s} ds + o(e^{\frac{\alpha}{2}t}).$$

One may check that $k_\alpha = 1$ and

$$A_{\alpha,1} = \frac{1}{\beta} \mathbf{v} \mathbf{u}^\top, \quad \text{so} \quad W(\alpha, 1) = \frac{W}{\beta} \mathbf{u}^\top.$$

$\rightsquigarrow$ the term coming from α is the one from Nerman's LLN

Asymptotic behaviour when $\partial\Lambda = \emptyset$

Theorem (Kolesko, K., Meiners)

Assume that ξ, φ satisfy second moment assumptions and that $\operatorname{Re}\lambda > \frac{\alpha}{2}$ for all $\lambda \in \Lambda$. Put

$$H_{\Lambda}(t) = \sum_{\lambda \in \Lambda} e^{\lambda t} \sum_{k=0}^{k_{\lambda}-1} W(\lambda, k+1) \int_{\mathbb{R}} \mathbb{E}[\varphi](s) \frac{(t-s)^k}{k!} e^{-\lambda s} ds.$$

Then, as $t \rightarrow \infty$,

$$e^{-\frac{\alpha}{2}t} (\mathcal{Z}_t^{\varphi} - H_{\Lambda}(t)) \xrightarrow{d} \sigma\left(\frac{W}{\beta}\right)^{1/2} \mathcal{N}$$

where $\mathcal{N}$ is a standard normal variable independent of W .

PROOF IDEAS

We have

$$W(\lambda) = \begin{pmatrix} W(\lambda, 1) \\ \vdots \\ W(\lambda, k_\lambda) \end{pmatrix} = \exp(\lambda, -t, k) \mathcal{Z}_t^{\varphi_\lambda + \chi_\lambda},$$

where

$$\exp(\lambda, s, k) := e^{\lambda s} \begin{pmatrix} 1 & s & \frac{s^2}{2} & \cdots & \frac{s^{k-1}}{(k-1)!} \\ & 1 & s & \cdots & \frac{s^{k-2}}{(k-2)!} \\ & & 1 & \ddots & \vdots \\ & & & \ddots & s \\ & & & & 1 \end{pmatrix}$$

and $\varphi_\lambda, \chi_\lambda$ are random, matrix-valued characteristics.

$$H_{\Lambda}(t)=\sum_{\lambda\in\Lambda}\sum_{k=0}^{k_{\lambda}-1}W(\lambda,k+1)\int_{\mathbb{R}}\mathbb{E}[\varphi](s)\frac{(t-s)^k}{k!}e^{\lambda(t-s)}\,\mathrm{d}s$$

$$\begin{aligned}
H_{\Lambda}(t) &= \sum_{\lambda \in \Lambda} \sum_{k=0}^{k_{\lambda}-1} W(\lambda, k+1) \int_{\mathbb{R}} \mathbb{E}[\varphi](s) \frac{(t-s)^k}{k!} e^{\lambda(t-s)} \, \mathrm{d}s \\
&= \sum_{\lambda \in \Lambda} \int_{\mathbb{R}} \mathbf{e}_1^{\top} \exp(\lambda, t-s, k_{\lambda}) W(\lambda) \mathbb{E}[\varphi](s) \mathrm{d}s
\end{aligned}$$

$$\begin{aligned}
H_{\Lambda}(t) &= \sum_{\lambda \in \Lambda} \sum_{k=0}^{k_{\lambda}-1} W(\lambda, k+1) \int_{\mathbb{R}} \mathbb{E}[\varphi](s) \frac{(t-s)^k}{k!} e^{\lambda(t-s)} \, ds \\
&= \sum_{\lambda \in \Lambda} \int_{\mathbb{R}} \mathbf{e}_1^{\top} \exp(\lambda, t-s, k_{\lambda}) W(\lambda) \mathbb{E}[\varphi](s) \, ds \\
&= \sum_{\lambda \in \Lambda} \int_{\mathbb{R}} \mathbf{e}_1^{\top} \exp(\lambda, t-s, k_{\lambda}) \exp(\lambda, -t, k_{\lambda}) \mathcal{Z}_t^{\varphi_{\lambda} + \chi_{\lambda}} \mathbb{E}[\varphi](s) \, ds
\end{aligned}$$

$$\begin{aligned}
H_{\Lambda}(t) &= \sum_{\lambda \in \Lambda} \sum_{k=0}^{k_{\lambda}-1} W(\lambda, k+1) \int_{\mathbb{R}} \mathbb{E}[\varphi](s) \frac{(t-s)^k}{k!} e^{\lambda(t-s)} \, ds \\
&= \sum_{\lambda \in \Lambda} \int_{\mathbb{R}} \mathbf{e}_1^{\top} \exp(\lambda, t-s, k_{\lambda}) W(\lambda) \mathbb{E}[\varphi](s) \, ds \\
&= \sum_{\lambda \in \Lambda} \int_{\mathbb{R}} \mathbf{e}_1^{\top} \exp(\lambda, t-s, k_{\lambda}) \exp(\lambda, -t, k_{\lambda}) \mathcal{Z}_t^{\varphi_{\lambda} + \chi_{\lambda}} \mathbb{E}[\varphi](s) \, ds \\
&= \sum_{\lambda \in \Lambda} \int_{\mathbb{R}} \mathbf{e}_1^{\top} \exp(\lambda, -s, k_{\lambda}) \mathcal{Z}_t^{\varphi_{\lambda} + \chi_{\lambda}} \mathbb{E}[\varphi](s) \, ds
\end{aligned}$$

$$\begin{aligned}
H_{\Lambda}(t) &= \sum_{\lambda \in \Lambda} \sum_{k=0}^{k_{\lambda}-1} W(\lambda, k+1) \int_{\mathbb{R}} \mathbb{E}[\varphi](s) \frac{(t-s)^k}{k!} e^{\lambda(t-s)} \mathrm{d}s \\
&= \sum_{\lambda \in \Lambda} \int_{\mathbb{R}} \mathbf{e}_1^{\top} \exp(\lambda, t-s, k_{\lambda}) W(\lambda) \mathbb{E}[\varphi](s) \mathrm{d}s \\
&= \sum_{\lambda \in \Lambda} \int_{\mathbb{R}} \mathbf{e}_1^{\top} \exp(\lambda, t-s, k_{\lambda}) \exp(\lambda, -t, k_{\lambda}) \mathcal{Z}_t^{\varphi_{\lambda} + \chi_{\lambda}} \mathbb{E}[\varphi](s) \mathrm{d}s \\
&= \sum_{\lambda \in \Lambda} \int_{\mathbb{R}} \mathbf{e}_1^{\top} \exp(\lambda, -s, k_{\lambda}) \mathcal{Z}_t^{\varphi_{\lambda} + \chi_{\lambda}} \mathbb{E}[\varphi](s) \mathrm{d}s = \mathcal{Z}_t^{\psi}
\end{aligned}$$

for the characteristic

$$\psi(t) = \sum_{\lambda \in \Lambda} \int_{\mathbb{R}} \mathbf{e}_1^{\top} \exp(\lambda, -s, k_{\lambda}) (\varphi_{\lambda}(t) + \chi_{\lambda}(t)) \mathbb{E}[\varphi](s) \mathrm{d}s.$$

Reduction I : slowly growing mean

We have

$$\mathcal{Z}_t^\varphi - H_\Lambda(t) = \mathcal{Z}_t^{\varphi-\psi}$$

Reduction I : slowly growing mean

We have

$$\mathcal{Z}_t^\varphi - H_\Lambda(t) = \mathcal{Z}_t^{\varphi-\psi}$$

with

$$\mathbb{E}[\mathcal{Z}_t^{\varphi-\psi}] = r(t) = o(e^{\frac{\alpha}{2}t})$$

Reduction I : slowly growing mean

We have

$$\mathcal{Z}_t^\varphi - H_\Lambda(t) = \mathcal{Z}_t^{\varphi-\psi}$$

with

$$\mathbb{E}[\mathcal{Z}_t^{\varphi-\psi}] = r(t) = o(e^{\frac{\alpha}{2}t})$$

and the goal is

$$e^{-\frac{\alpha}{2}t} \mathcal{Z}_t^{\varphi-\psi} \xrightarrow{d} \sigma\left(\frac{W}{\beta}\right)^{1/2} \mathcal{N}.$$

Reduction II : centered characteristic

Consider $\mathcal{Z}_t^\phi$ such that $\mathbb{E}[\mathcal{Z}_t^\phi] = o(e^{\frac{\alpha}{2}t})$.

Reduction II : centered characteristic

Consider $\mathcal{Z}_t^\phi$ such that $\mathbb{E}[\mathcal{Z}_t^\phi] = o(e^{\frac{\alpha}{2}t})$. Then there exists a **centered** characteristic χ such that

$$\mathcal{Z}_t^\phi = \mathcal{Z}_t^\chi + \mathbf{e}_{\tau(\emptyset)}^\top \mathbb{E}[\mathcal{Z}_t^\phi]$$

Reduction II : centered characteristic

Consider $\mathcal{Z}_t^\phi$ such that $\mathbb{E}[\mathcal{Z}_t^\phi] = o(e^{\frac{\alpha}{2}t})$. Then there exists a **centered** characteristic χ such that

$$\mathcal{Z}_t^\phi = \mathcal{Z}_t^\chi + \mathbf{e}_{\tau(\emptyset)}^\top \mathbb{E}[\mathcal{Z}_t^\phi] = \mathcal{Z}_t^\chi + o(e^{\frac{\alpha}{2}t}).$$

If ξ, ϕ satisfy second moment assumptions, then so does χ .

Reduction II : centered characteristic

Consider $\mathcal{Z}_t^\phi$ such that $\mathbb{E}[\mathcal{Z}_t^\phi] = o(e^{\frac{\alpha}{2}t})$. Then there exists a **centered** characteristic χ such that

$$\mathcal{Z}_t^\phi = \mathcal{Z}_t^\chi + \mathbf{e}_{\tau(\emptyset)}^\top \mathbb{E}[\mathcal{Z}_t^\phi] = \mathcal{Z}_t^\chi + o(e^{\frac{\alpha}{2}t}).$$

If ξ, ϕ satisfy second moment assumptions, then so does χ .

— here $\chi = \phi - \mathbb{E}\phi + \xi * U * \mathbb{E}\phi - \mu * U * \mathbb{E}\phi$.

Centered characteristic

We have

$$\mathcal{Z}_t^\chi = \sum_u \mathbf{e}_{\tau(u)}^\top \chi_u(t - S(u))$$

with $\mathbb{E}\chi \equiv 0$.

Centered characteristic

We have

$$\mathcal{Z}_t^\chi = \sum_u \mathbf{e}_{\tau(u)}^\top \chi_u(t - S(u))$$

with $\mathbb{E}\chi \equiv 0$.

Enumerate all individuals in the genealogical tree

Centered characteristic

We have

$$\mathcal{Z}_t^\chi = \sum_u \mathbf{e}_{\tau(u)}^\top \chi_u(t - S(u))$$

with $\mathbb{E}\chi \equiv 0$.

Enumerate all individuals in the genealogical tree and write

$$M_n(t) = \sum_{k=1}^n \mathbf{e}_{\tau(u_k)}^\top \chi_{u_k}(t - S(u_k)), \quad \mathcal{Z}_t^\chi = \lim_{n \rightarrow \infty} M_n(t).$$

Centered characteristic

We have

$$\mathcal{Z}_t^\chi = \sum_u \mathbf{e}_{\tau(u)}^\top \chi_u(t - S(u))$$

with $\mathbb{E}\chi \equiv 0$.

Enumerate all individuals in the genealogical tree and write

$$M_n(t) = \sum_{k=1}^n \mathbf{e}_{\tau(u_k)}^\top \chi_{u_k}(t - S(u_k)), \quad \mathcal{Z}_t^\chi = \lim_{n \rightarrow \infty} M_n(t).$$

$(M_n(t))_{n \in \mathbb{N}}$ is a martingale: with $\mathcal{G}_n = \sigma(\text{lifespaces of } u_1, \dots, u_n)$,

$$\mathbb{E}[\chi_{u_n}(t - S(u_n)) | \mathcal{G}_{n-1}] = \mathbb{E}[\chi](t - S(u_n)) = 0.$$

Centered characteristic

We have

$$\mathcal{Z}_t^\chi = \sum_u \mathbf{e}_{\tau(u)}^\top \chi_u(t - S(u))$$

with $\mathbb{E}\chi \equiv 0$.

Enumerate all individuals in the genealogical tree and write

$$M_n(t) = \sum_{k=1}^n \mathbf{e}_{\tau(u_k)}^\top \chi_{u_k}(t - S(u_k)), \quad \mathcal{Z}_t^\chi = \lim_{n \rightarrow \infty} M_n(t).$$

$(M_n(t))_{n \in \mathbb{N}}$ is a martingale: with $\mathcal{G}_n = \sigma(\text{lifespaces of } u_1, \dots, u_n)$,

$$\mathbb{E}[\chi_{u_n}(t - S(u_n)) | \mathcal{G}_{n-1}] = \mathbb{E}[\chi](t - S(u_n)) = 0.$$

$\rightsquigarrow$ use Martingale CLT!

THANK YOU FOR YOUR ATTENTION!

What if $\partial\Lambda \neq \emptyset$?

Put

$$H_{\Lambda}(t) = \sum_{\operatorname{Re}\Lambda > \frac{\alpha}{2}} e^{\lambda t} \sum_{k=0}^{k_{\lambda}-1} W(\lambda, k+1) \int_{\mathbb{R}} \mathbb{E}[\varphi](s) \frac{(t-s)^k}{k!} e^{-\lambda s} \mathrm{d}s,$$

$$H_{\partial\Lambda}(t) = \sum_{\operatorname{Re}\lambda = \frac{\alpha}{2}} e^{\lambda t} \sum_{k=0}^{k_{\lambda}-1} \mathbf{e}_{\tau(\emptyset)}^{\top} A_{\lambda, k+1} \int_{\mathbb{R}} \mathbb{E}[\varphi](s) \frac{(t-s)^k}{k!} e^{-\lambda s} \mathrm{d}s.$$

If the terms coming from $\lambda \in \partial\Lambda$ are non-trivial, then, for some $n \in \mathbb{N}$,

$$\frac{e^{-\frac{\alpha}{2}t}}{\sqrt{t^{2n+1}}} (\mathcal{Z}_t^{\varphi} - H_{\Lambda}(t) - H_{\partial\Lambda}(t)) \xrightarrow{\mathrm{d}} \sigma \left(\frac{W}{\beta} \right)^{1/2} \mathcal{N}$$

and σ is given **solely** by the terms coming from $\lambda \in \partial\Lambda$.